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Assessment of organ and size-specific effective doses from cone beam CT (CBCT) in image-guided radiotherapy (IGRT) based on body mass index (BMI)

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ABSTRACT

The ability of radiation therapy to deliver radiation doses accurately to tumour targets, while sparing normal tissues, has advanced remarkably over recent decades. A development that has been key to achieving this improvement has been the application of image-guidance that facilitates more precise delivery of the prescribed dose to the treatment site. A recent international survey showed that kV cone beam CT (CBCT) is the imaging modality used most widely for image guided radiation therapy (IGRT) procedures. However, IGRT will deliver additional radiation doses to patients that should be considered. The aims of this study were to (1) estimate organ and size-specific effective doses resulting from kV-CBCT chest and pelvis scans based on body mass index (BMI) of the patient, and (2) investigate the influence of patient size on imaging dose. A validated Monte Carlo model developed with BEAMnrc/EGSnrc was employed to simulate kV spectra generated by Varian on-board imaging (OBI) system. Organ and size-specific effective doses were assessed for adult phantoms in the National Cancer Institute (NCI) phantom library using DOSXYZnrc/EGSnrc code. The library has been reconstructed from images of CT patients, and contains 100 male and 93 female phantoms of a wide range of sizes. The phantoms were grouped into six categories based on body mass index (BMI) as classified by the world health organization. For all sizes, average doses to organs that lay fully or partially inside the field of the chest and pelvic scans were in the range of (0.20–3.06 mGy/100 mAs) and (0.17–4.47 mGy/100 mAs), and size-specific effective doses were (0.63 – 1.78 mSv/100 mAs) and (0.30 – 1.17 mSv/100 mAs), respectively. Patient size played a significant role in determining the imaging doses resulting from both scans. In general, organ and effective doses of chest and pelvic scans of thin patients were about 2–3 times larger than those of obese patients, with the impact of patient size on pelvis doses being slightly higher than on chest ones. Results of this study suggests that patient size should be taken into consideration when estimation and optimization of the imaging doses resulting from CBCT. Although development of patient size-specific CBCT protocols is a practical approach that can account for patient size, it has not been implemented worldwide. Results from this investigation might be used as a simple approach to obtain an estimation of the imaging doses for patients of various sizes based on BMI.

1. Introduction

Image guidance has become an essential component of the majority of radiotherapy treatments worldwide in the last decade. Image guided radiation therapy (IGRT) helps to improve the outcome of treatment significantly by delivering radiation therapy to the planned target (i.e. tumour) more precisely. As with other treatment procedures, IGRT comes with some side effects in that it delivers additional doses to healthy organs and tissues of patients. Although imaging doses involved in IGRT procedures are substantially lower than those used for therapy, cumulative doses resulting from, for example, the daily usage of an imaging modality, may no longer be negligible (Ding et al., 2018; Martin and Barnard, 2022; Rehani et al., 2020). Some protocols require the acquisition of a scan on a daily basis to ensure that the fraction dose is delivered accurately (Posiewnik and Piotrowski, 2019). These protocols are usually applied when high fractional doses are planned to be

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delivered, such as those used for stereotactic ablative body radiotherapy (SABR), where target doses of 10 Gy per fraction might be given to patients for short periods of time.

Several imaging modalities may be used for IGRT procedures, some of which are acquired in the MV energy range such as electronic portal imaging devices (EPIDs) and MV cone beam computed tomography (CBCT), while other modalities are based on kV energies such as radiography, kV-CBCT, and fluoroscopy (Murphy et al., 2007; Ding et al., 2018). In addition, options using non-ionizing radiations such as magnetic resonance imaging (MRI), ultrasound and optical surface guidance are used, where these are available. A recent international survey has shown that kV-CBCT is used widely around the globe (Martin et al., 2021). kV-CBCT is found to be superior because it provides 3D images of good resolution, particularly for bone structures, which are used for patient positioning through comparison with CT images acquired for treatment planning.

Currently, two kV CBCT systems integrated into linear accelerators (LINACs) are widely used for kV-CBCT scans during radiotherapy treatment, these are the Varian on-board imaging (OBI) and Elekta x-ray volume imaging (XVI) systems. Imaging organ doses resulting from these systems are investigated in many studies in the literature, most of which are summarized in two reviews (Alaei and Spezi, 2015; Posiewnik and Piotrowski, 2019). Generally, the patient organ and tissue doses are assessed by using Monte Carlo simulations on CT images of patients or on reference computational human phantoms such as those provided by the International Commission on Radiological Protection (ICRP, 2009). The patient dose could also be assessed using physical measurements on human-like phantoms such as anthropomorphic phantoms with an appropriate dosimetry system (Alaei and Spezi, 2015). The main limitation of these studies for the evaluation of doses for individual patients is that most were conducted on phantoms of reference sizes, and some were based on one gender. This gives an estimation of the average doses, but may lead to under or overestimation of doses to patients of varying build, especially underweight or obese patients. Therefore, the aims of this study were (1) to assess organ and size-specific effective doses of kV-CBCT utilized in IGRT procedures for a wide range of patient sizes (Martin et al., 2020), and (2) to investigate the influence of the patient size on the imaging dose as compared to the average (i.e. reference) size. The size-specific effective dose combines organ and tissue doses derived from a phantom of a specific size, usually chosen to represent an individual patient, using a similar formulation and tissue weighting factors to those used to derive effective dose (Martin et al., 2020), whereas the effective dose is based on reference standard size phantoms.

2. Materials and methods

2.1. Computational human phantoms

A total of 193 hybrid computational human adult phantoms of a wide range of sizes were used in this study. Those phantoms are part of a large library developed by the National Cancer Institute (NCI) of the US and University of Florida (Lee et al., 2010; Geyer et al., 2014; NCI, 2023). More than 1000 CT scans of patients were used to establish those phantoms to ensure coverage of the full patient population. The library contains 193 adult phantoms as well as 158 pediatric phantoms, but the latter were not included in this study. The adult phantoms were 100 male and 93 female phantoms with heights and weight of 160–190 cm and 50–125 kg, and 150–175 cm and 40–120 kg, respectively. In order to investigate the influence of patient size on organ and size-specific effective doses, the phantoms were divided based on the body mass index (BMI) into six classes listed in Table 1 as categorized by the world health organization (WHO, 2022).

2.2. Monte Carlo simulations

Organ and size-specific effective doses were assessed for the adult phantoms using Monte Carlo (MC) simulations, which were performed in two separate steps. The first step involved simulating the kV source before interaction of photons with the patient body, while acquisition of kV-CBCT scans was simulated in the second step. A previously validated MC model of the OBI kV system integrated in a Truebeam linac was employed (Abuhaimed et al., 2014, 2015). BEAMnrc code, which is based on the EGSnrc system, was used to simulate the kV beam with a tube potential 120 kV (Rogers et al., 1995; Kawrakow, 2000; Kawrakow I. et al., 2000). A field size of 22 cm (longitudinal to the patient) x 28.1 cm (lateral), which is similar to the default protocol, was used for all simulations. The default protocol provided by Varian comes with a single field size for each scan protocol, whereas Elekta provides the XVI system with several field sizes, from which the user selects. A total of 1.0×10^{10} particles were run to simulate the beam passing through all the system components. In order to improve efficiency of the simulations, the directional bremsstrahlung splitting (DBS) was set with the split value at 2.0×10^4 (Kawrakow et al., 2004). The electron and photon transport cutoff energies were set to ECUT = 0.516 MeV and PCUT = 0.001, respectively (Rogers et al., 2021). The resulting spectrum was stored in a file known as phase space, which was transferred to the second step (Walters et al., 2021).

DOSXYZnrc code was used to simulate acquisition of kV-CBCT scans, from which organ and size-specific effective doses was assessed on the adult phantoms (Kawrakow and Walters, 2006; Mainegra-Hing and Kawrakow, 2006; Walters et al., 2002). A MATLAB code developed in the house was used to transfer the format of the NCI phantoms to DOSXYZnrc format (Walters et al., 2021). The phase space file obtained in the first step with BEAMnrc was used as a source to run chest and pelvis kV-CBCT scans. Two simulations were performed for each phantom by rotating the phase space file 360° around the scan areas, one at the middle of the lung for the chest and the second at the urinary bladder for the pelvis scan. Results of the simulations were stored in 3D dose files, which were analysed subsequently by a MATLAB code written to extract the mean dose to each organ and hence to assess size-specific effective dose. The method used in this study was similar to those applied in previous studies (Martin et al., 2016; Abuhaimed et al., 2018; Abuhaimed and Martin, 2018). All simulations were performed in the supercomputing laboratory (Shaheen) at King Abdullah University of Science and Technology (KAUST).

Table 1
The classifications used to categorize the adult phantoms based on the BMI.

Classes	Nutritional status	BMI (kg/m^2)	No. of phantoms
C1	Underweight	Below 18.5	10
C2	Normal weight	18.5–24.9	49
C3	Pre-obesity	25.0–29.9	37
C4	Obesity class I	30.0–34.9	36
C5	Obesity class II	35.0–39.9	33
C6	Obesity class III	Above 40	28
Total			193

3. Results and discussion

Tables 2 and 3 list organ and size-specific effective doses of chest and pelvis scans for all sizes classified based on the BMI. Only organs, which were fully or partially inside the scan field, are considered. For all scans investigated, the patient size was found to play a significant role in determining organ and size-specific effective doses. For the chest scan, the average lung dose of the lowest BMI group was (3.53 mGy/100 mAs), and this declined to about half (1.71 mGy/100 mAs) in the largest BMI group (Table 2). Large differences in organ doses between the BMI classes were also found for organs such as heart, thymus, oesophagus, and thyroid. The breast dose did not change markedly with patient size in these results, because the breast is a superficial organ and is less affected by patient body size, However, in females, it is likely to vary

Table 2
Average organ and effective doses resulting from chest kV-CBCT scan of adult phantoms categorized into six classifications based on the BMI (Table 1).

Chest	Organ dose (mGy/100 mAs)						
	C1	C2	C3	C4	C5	C6	All
RBM	1.42	1.19	1.02	0.91	0.84	0.82	1.00
Lung	3.53	2.89	2.34	2.02	1.81	1.71	2.30
Stomach	1.05	0.83	0.69	0.63	0.60	0.63	0.71
Breast	3.24	2.87	2.55	2.30	2.15	2.07	2.48
Adrenals	0.65	0.52	0.45	0.39	0.37	0.38	0.44
Gall bladder	0.60	0.46	0.40	0.35	0.35	0.38	0.40
Heart	4.48	3.72	3.04	2.51	2.22	2.00	2.90
Kidneys	0.26	0.21	0.20	0.18	0.18	0.19	0.20
Pancreas	0.34	0.27	0.24	0.22	0.22	0.24	0.24
Spleen	0.82	0.67	0.58	0.54	0.53	0.56	0.60
Thymus	4.54	3.90	3.31	2.74	2.36	2.01	3.06
Oesophagus	3.30	2.62	2.07	1.69	1.46	1.32	1.99
Liver	1.56	1.16	0.95	0.83	0.78	0.80	0.96
Thyroid	1.67	1.33	1.28	1.13	1.12	1.08	1.23
Skin	0.60	0.52	0.47	0.44	0.43	0.41	0.47
Effective dose (mSv/100 mAs)	1.53	1.27	1.08	0.95	0.88	0.85	1.06

RBM: red bone marrow.

Table 3
Average organ and effective doses resulting from pelvis kV-CBCT scan of adult phantoms categorized into six classifications based on the BMI (Table 1).

Pelvis	Organ dose (mGy/100 mAs)						
	C1	C2	C3	C4	C5	C6	All
RBM	1.60	1.40	1.18	1.05	0.92	0.78	1.13
Colon	0.82	0.68	0.61	0.55	0.51	0.47	0.59
Small intestine	1.43	1.24	1.01	0.83	0.71	0.58	0.94
Kidneys	0.21	0.20	0.17	0.16	0.15	0.14	0.17
Prostate	4.58	3.92	3.57	3.16	2.78	2.23	3.37
Uterus	3.50	2.90	2.25	1.80	1.46	1.18	2.05
Ovaries	3.82	3.18	2.46	1.97	1.63	1.32	2.26
Testes	5.48	4.70	4.57	4.38	4.28	3.60	4.47
Urinary bladder	4.36	3.73	3.07	2.54	2.14	1.66	2.84
Skin	0.66	0.57	0.52	0.48	0.46	0.44	0.51
Effective dose (mSv/100 mAs)	0.92	0.81	0.71	0.63	0.56	0.43	0.66

RBM: red bone marrow.

with breast size, but this was not investigated here. Large differences in organ dose between the BMI classes were also seen in organs such as the stomach and liver that were not fully inside the scan field. This is likely to result mainly from the thickness of the overlying fatty tissues, which increased with patient weight, but there will be some variation with height, where a greater proportion of an organ lies within the scan field for shorter patients. As a result of the large differences in organ doses caused by the patient size, specifically organs of higher tissue weighting factors, the size-specific effective dose per unit exposure varied significantly among the BMI classes. The average size-specific effective dose for obese patients (C6) was (0.85 mSv/100 mAs), which was doubled to (1.53 mSv/100 mAs) for thin patients (C1).

The influence of patient size on organ and size-specific effective doses was also found to be significant for pelvis scans (Table 3). The differences in organ doses from pelvis scans were slightly larger than those for chest scans for certain organs. For example, the average dose to the urinary bladder of thin patients (4.36 mGy/100 mAs) was about three times larger than that for obese patients (1.66 mGy/100 mAs). Doses to the prostate, uterus, ovaries, and testes were also affected remarkably by the size of the patient. This influence was also extended to organs that were not fully exposed during the scans such as colon and small intestine. However, since tissue weighting factors for the gonads and urinary bladder are smaller as compared to organs in the chest area, the impact of patient size on the size-specific effective dose was less for pelvis scans, and only increased by about 0.5 mSv/100 mAs for thin patients.

Doses listed in Tables 2 and 3 are the average doses for each class. Figs. 1–3 show more details about organ and size-specific effective dose within each class. It can be clearly seen that the influence of patient size was significant not only between the BMI classes, but also among patients of the same class. For example, thyroid doses for the pre-obesity class (C3) resulting from chest scans varied by more than 2 mGy/100 mAs among patients of the same class. This is due to differences in patient height, because when a standard field size is used, a greater proportion of the thyroid, which lies on the periphery of the chest field, will be included in scans of shorter patients (Martin and Abuhaimed, 2022). Such differences were also seen for colon and small intestine doses in pelvis scan. As would be expected, organ and size-specific effective doses decreased with increasing patient size, except for the thyroid, which due its superficial position, was not affected significantly by the BMI of the patient, but It will be affected by height.

Results found in this study suggest that patient size-specific CBCT protocols should be considered in order to optimize radiological protection and to give more precision in assessment of the patient organ doses. AAPM TG-180 describes two methods, through which imaging dose for a specific patient can be accounted for as part of the treatment dose: (1) patient-specific imaging dose calculations and (2) non-patient-specific imaging dose calculations (Ding et al., 2018). Dose assessment using the first method is based on using treatment planning systems (TPSs) or a model-based dose calculation algorithm before treatments. While the first method can be simply applied for MV imaging modalities, use of TPSs to account for imaging doses in the kV energy range would require additional modelling and commissioning of kV TPS algorithms. Several studies have been conducted to account for doses from kV imaging using TPSs or model-based dose calculation algorithms (Alaei and Spezi, 2012; Dzierma et al., 2014; Pawlowski and Ding, 2014). However, those studies reveal that there are challenges with uncertainties in doses to some body structures, which can be as high as ±300%. As a result, this method is not widely available at the present time, but it may be possible to include in TPS algorithms in the future. The second method is based on the fact that the dose variations between different studies of kV-CBCT conducted under different conditions have been found to be relatively small (Ding et al., 2018). This has led to suggestions that a simpler approach could be appropriate with the development of look–up tables using various studies that covered different scan protocols, patient sizes, and kV imaging systems.

Recently, several studies suggest development of patient size-specific CBCT protocols. Ordóñez-sanz et al. showed that patients undergoing radiotherapy pelvis treatment can be divided into four groups small (S), medium (M), large (L) and extra-large (XL) based on the $CTDI_{vol}$ of the CT scans that were taken for treatment planning (Ordóñez-Sanz et al., 2021). Subsequently, specific CBCT protocols were established, one for each group, with different scan parameters from the default protocols provided by manufacturers. Use of the protocols developed resulted in dose reductions in the range of 19–83% for the first three groups, while the imaging dose of the XL group increased by up to 39% as compared to the default protocols. Similarly, Khan et al. developed various patient size-specific CBCT protocols that were different from the default protocols and covered head and neck, thorax and abdomen/pelvic scans (Khan et al., 2022). They divided the patients into three groups (S, M, and L) based on the diameter of the patient body. A total of 1185 patients were scanned with the new protocols, and results showed that the imaging dose was reduced for 52% of the patients, remained the same for 37%, and increased for remainder (12%). Moreover, it has been shown that increasing size of the scan field of kV-CBCT along the patient body by 2 cm can increase the patient dose by 12% for chest scan and by 13% for pelvis scan (Martin and Abuhaimed, 2022). Therefore, it has been suggested that optimization of the size of the scan field based on patient size should be considered. Reduction in the number of radiographic projections was also suggested to reduce the patient dose while

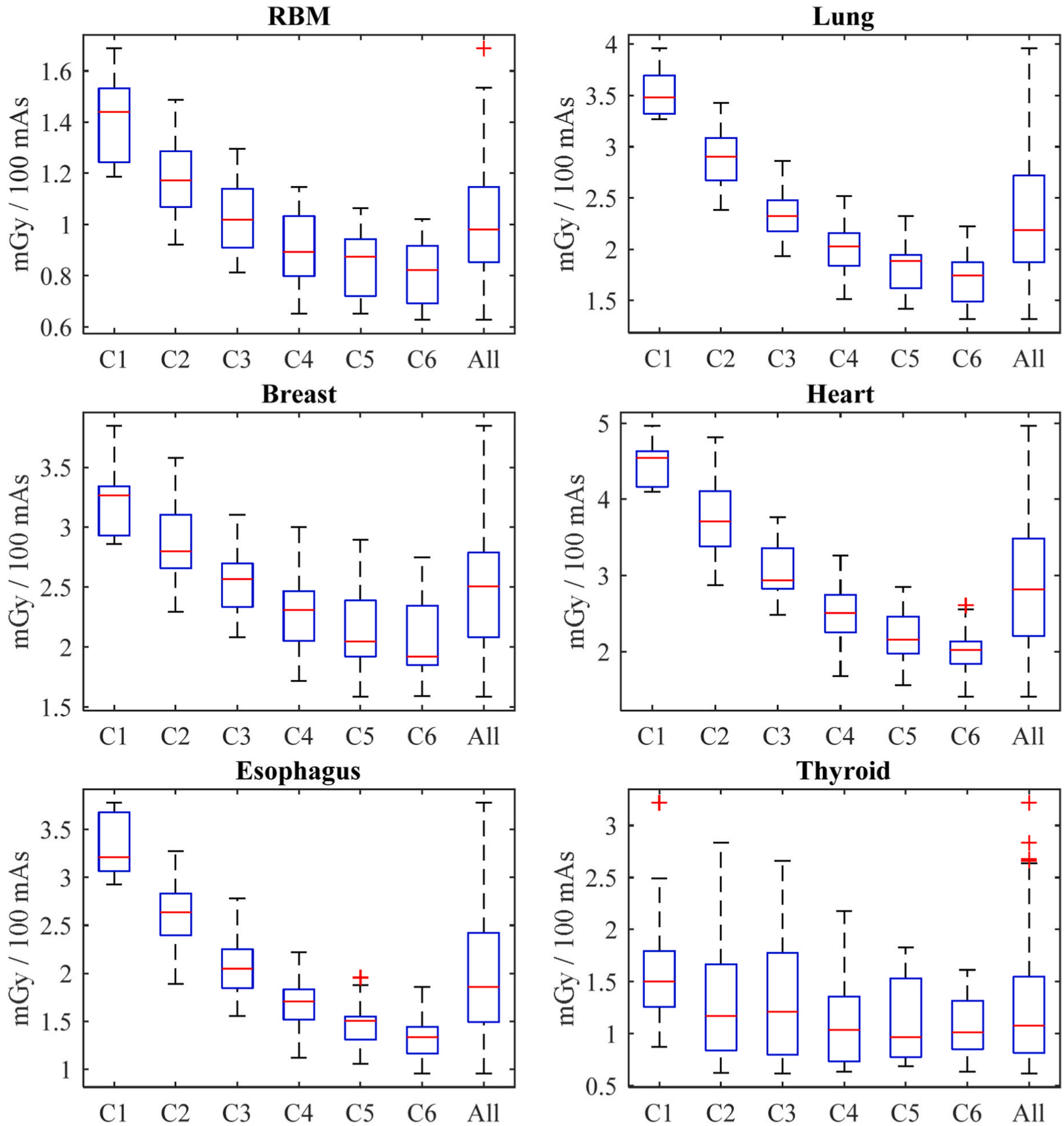

Fig. 1. Box and whisker plot of doses to some organs resulting from chest scan for each BMI class (Table 1).

maintaining the image quality (Al-Kabkabi et al., 2022).

Although the imaging dose resulting from kV-CBCT is lower than that used for treatment, optimization should be considered in order to minimize doses to normal tissues. It has been shown that the additional dose from imaging may lead to variations in tumour response and the risk of morbidity if imaging doses reached 5% of the total prescribed dose (Ding et al., 2018). Although patient size-specific CBCT protocols have been shown to be a practical approach to reduce any unnecessary additional dose, it has not been implemented in many places worldwide, and the default protocols are in common use. Therefore, the non-patient-specific imaging dose calculation method could provide a practical approach for evaluating and including the dose in treatment prescriptions. Results of this study, are non-patient-specific, but show that a simple approach based on the BMI of a patient could be used to estimate organ doses that a patient will receive during the treatment. This should provide clinical practitioners with a reasonable estimation of the imaging dose with an acceptable level of accuracy. This could then be used to ensure that the sums of doses from therapy and imaging of organs at risk remain below the relevant threshold tolerances.

The main limitation of this study is that it has been conducted with MC simulations on phantoms reconstructed from real CT images. This is expected to result in some dose uncertainties due to the accuracy of the simulations, but also variations due to organ positions within individual phantoms. However, it is expected that the uncertainty will not exceed ±20%, and this is acceptable for estimation of imaging doses (Ding et al., 2018). Moreover, only adult patients were considered, as scan protocols of pediatric patients are not widely available.

4. Conclusions

MC simulations were used to assess imaging doses resulting from kV-CBCT chest and pelvis scans utilized for IGRT procedures using the NCI phantom library of a wide range of sizes. The patient BMI influenced both organ and size-specific effective doses remarkably, where dose variations between the different sizes reached to 2–3 times. Such large variations suggest that establishment of patient size-specific CBCT

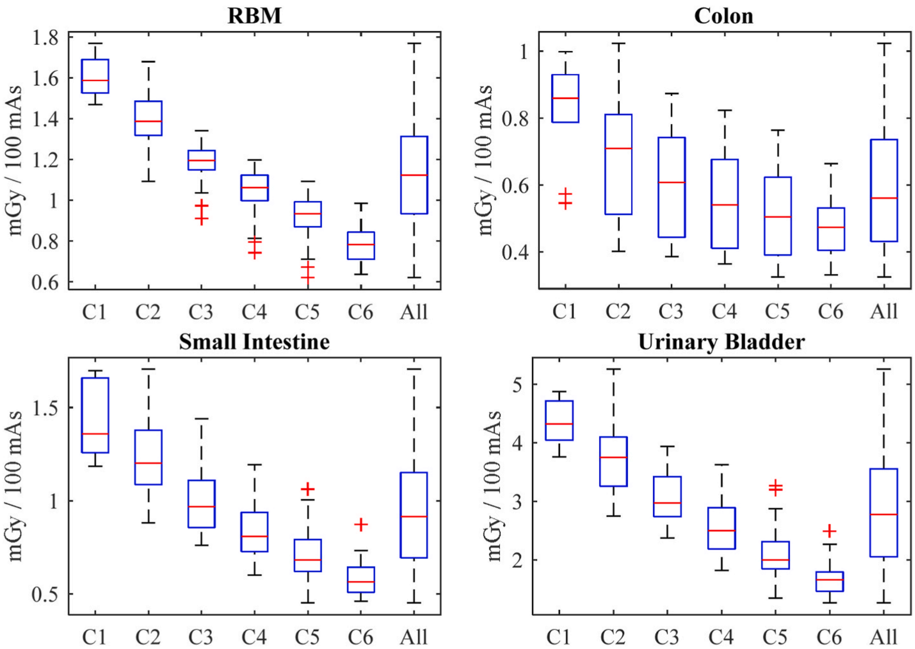

Fig. 2. Box and whisker plot of doses to some organs resulting from pelvis scan for each BMI class (Table 1).

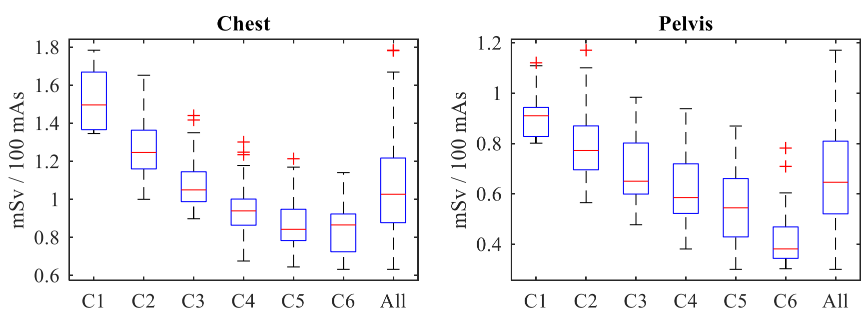

Fig. 3. Box and whisker plot of size-specific effective doses resulting from chest and pelvis scans for each BMI class (Table 1).

protocols that differ from the default scan protocols provided by manufactures should be considered when optimizing radiological protection for imaging procedures to account for the impact of patient size. Doses to normal tissues surrounding the tumour target, especially those to organs at risk, could be evaluated using a non-patient-specific method that is based on published studies to establish look–up tables. These could be used to assess patient organ doses with an acceptable level of accuracy, to enable the exposures to be taken into account in maintaining doses to organs at risk below tolerance levels, and accounting for the exposure of normal tissues during imaging.

Author statement

Abdullah Abuhaimed: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - Original Draft. Colin J. Martin: Conceptualization, Methodology, Investigation, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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